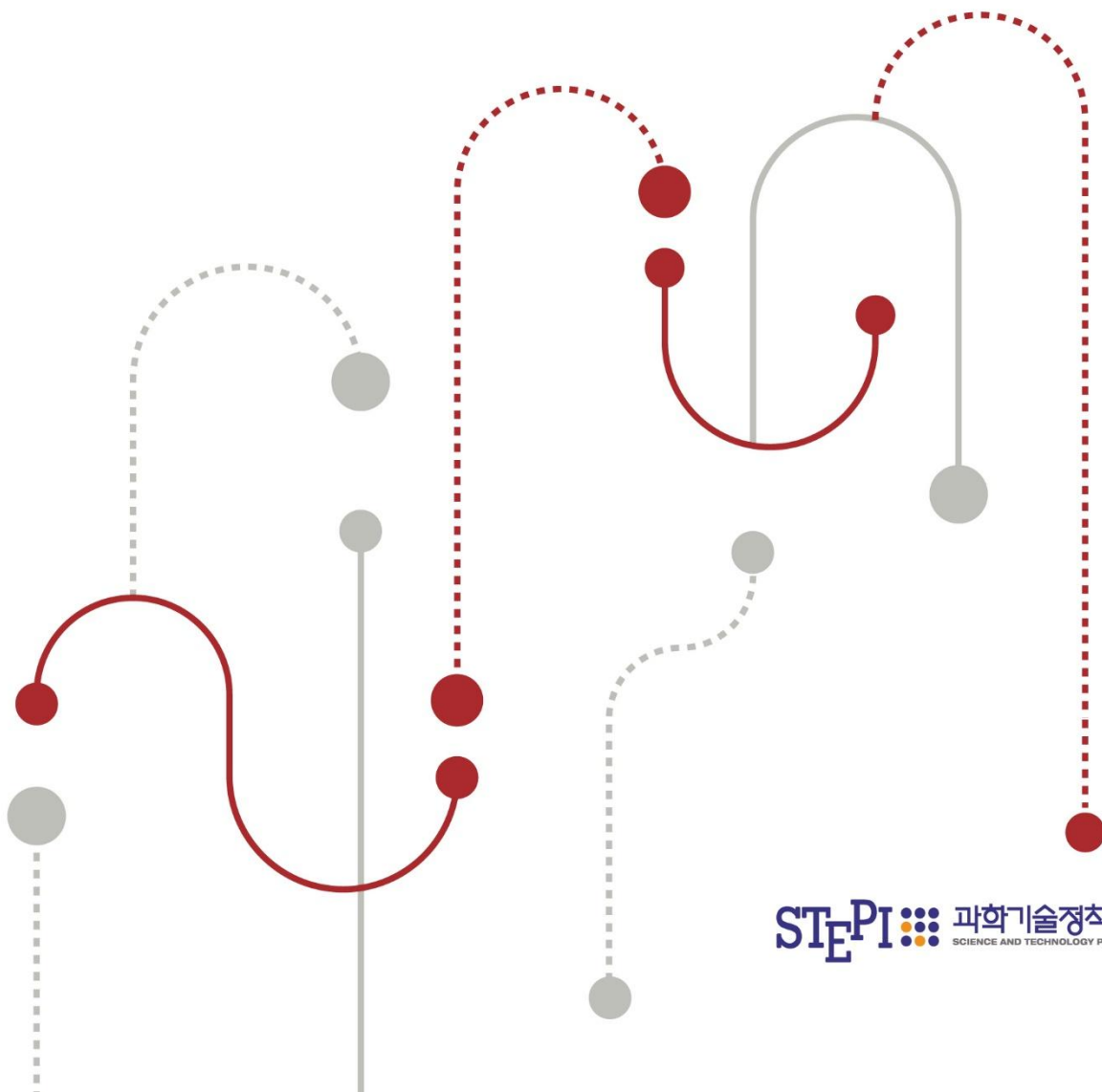


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Global Deep-Tech Startup Support Policies and Implications for Korea



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SUMMARY

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■ Concept of deep-tech and the need to support deep-tech startups

- Deep-tech startups refer to companies that build barriers to technological imitation based on fundamental scientific and engineering technologies and translate laboratory-level knowledge into market-applicable solutions
- Deep tech is characterized by the interdependence among technology, markets, and regulation, requiring an integrated pipeline that links research, prototyping, pilot projects, standards and certification, and early demand creation, particularly within the Technology Readiness Level (TRL) 3–7 range
 - After technological validity is demonstrated, investment demand increases sharply, and large-scale capital financing in later funding rounds becomes a decisive factor for growth
 - This underscores the need for a combination of full-cycle support and growth capital, as well as the design of qualification linkages and bridge mechanisms between stage

■ Policy trends for deep-tech startups in U.S. and Europe

- United States has maintained technological leadership through a long-term, strategic R&D approach linked to defense and national security, with recent emphasis on supply chain stabilization
- European countries are systematically fostering deep-tech ecosystems through government-led large-scale fund formation and long-term strategic planning based on their respective national strengths

- (United Kingdom) Through the “UK Innovation Strategy,” seven strategic technologies have been identified, and the “Invest 2035” plan focuses on intensive investment in high-growth sectors
- (France) Centered on the public investment bank Bpifrance, the “Deep Tech Plan” and “France 2030” strategy aim to create 500 deep tech startups annually and nurture unicorn companies
- (Germany) More than EUR 10 billion in policy funds, including the “Deep Tech Future Fund” and the “Climate Fund(DTCF),” have been established to address funding gaps at the growth stage

■ Policy trends for deep-tech startups by ministry in Korea

- (Ministry of Science and ICT) Promotion of startups based on fundamental technologies and the development of regionally anchored R&D valleys
 - Through programs such as the “Deep Science Startup Promotion Program” and the “Deep Tech Scale-up Valley Development Program,” the ministry supports early-stage startups in high-difficulty advanced technology fields and fosters regionally specialized technology commercialization clusters
- (Ministry of Trade, Industry and Energy) Strengthening competitiveness in strategic industries and reinforcing industrial value chains
 - Through initiatives such as the “National Advanced Industry Technology Innovation Loan Program” and the “Mega-city Cooperation Advanced Industry Development Support Program,” the ministry provides policy financing to promote technology commercialization and encourages joint growth of regionally specialized industries
- (Ministry of SMEs and Startups) Selection of promising firms through private-sector participation and a public-private cooperation strategy to foster super-gap enterprises
 - Through “Deep Tech TIPS” and the “Super Gap 1000+ Project,” the ministry seeks to enhance private-sector investment and incubation capabilities based on market

demand and provide phased support to nurture leading innovative firms in deep tech sectors

■ Policy recommendations for supporting Korean deep-tech startups

Task 1. Strengthening the strategic orientation of deep-tech centered startup support

- Establishing an integrated national control tower for deep-tech startup support and formulation of a systematic development strategy
 - Integration of fragmented, ministry-specific startup support policies and establishment of a pan-government policy control tower to enhance national innovation competitiveness
 - Establishment of core objectives, performance indicators, and phased implementation strategies for nurturing and supporting deep-tech startups at the national level
- Linking national strategic technology innovation with startup support policies
 - Enhancing organic linkages between innovation support policies for national core strategic technologies and startup support policies, and developing differentiated innovation frameworks by strategic technology domain
- Promoting demand- and problem-driven deep-tech startup development based on technology convergence
 - Promoting technology convergence from a market perspective through a demand- and problem-based approach and design of market-driven, project-based deep-tech support programs

Task 2. Strengthening the government's role as a facilitator of deep-tech innovation commercialization

- Expansion of large-scale public investment for deep-tech startup growth
 - Expansion of government public investment to support deep-tech startup growth and improvement of investment efficiency through reforms in public fund management systems
- Regulatory reform to promote demonstration and commercialization of deep-tech innovation outcomes
 - Introduction of tailored regulatory reform systems to reduce regulatory uncertainty in deep-tech sectors and identify key sector-specific regulatory bottlenecks
 - Promotion of effective demonstration and commercialization based on technological characteristics through integrated operation of fragmented regulatory sandbox programs and enhancement of follow-up support
- Strengthening the government's role as an innovation demander through expanded public procurement
 - Strengthening efforts to create early viable markets by expanding public procurement opportunities for technologies and services developed by innovative deep-tech startups with high market potential
 - Advancement of a Korean-style SBIR/STTR program to strengthen linkages between SME/startup R&D outcomes and government innovation procurement demand

Task 3. Building a private-sector-led deep-tech investment and incubation ecosystem for deep-tech startups

- Establishment of customized, private-sector-led investment and incubation support systems tailored to each stage of startup growth
 - Expansion of entrepreneurial capital and innovative finance supply based on the creativity and autonomy of private investment markets
 - Development of models linking private investment and policy support tailored to each stage of startup growth
- Enhancing deep-tech investment expertise of ACs/VCs and strengthening private-sector leadership in government startup support programs
 - Provision of incentives to activate private deep-tech investment by ACs/VCs and improvement of investment efficiency through the development of domain-specialized ACs/VCs
 - Enhancement of autonomy and flexibility of private operators in public- private cooperative startup support programs such as TIPS
- Expansion of strategic investment by large and mid-sized enterprises and activation of M&A and IPO markets
 - Regulatory relaxation and innovation to strengthen the role of large and mid-sized enterprises as strategic investors, including through CVCs
 - Provision of specialized incentives to activate M&A markets and institutional reforms to promote IPOs

Task 4. Enhancing the entrepreneurship of innovation actors and global openness

- Expansion of incentives to strengthen the entrepreneurial role of universities and government-funded research institutes
 - Strengthening deep tech commercialization support capabilities of TLOs by technology and region
 - Provision of specialized support and incentives to promote startup creation by deep-tech researchers at universities and government-funded research institutes
- Cultivation of startup talent based on innovation leadership and establishment of a “Korean Young Innovators” role model
 - Creation of institutional environments to nurture domestic and international innovation talent and facilitate their inflow into the startup ecosystem
 - Foster “Korean Young Innovators” equipped with entrepreneurial capabilities centered on problem recognition, technology application, and global orientation, and development
- Strengthening globalization of innovation actors and expanding openness and connectivity of the domestic innovation ecosystem
 - Operation of global leading R&D–commercialization linkage projects through collaboration among universities, government-funded research institutes, and large enterprises
 - Establishment of global standards in the domestic startup ecosystem through institutional reforms such as enhanced labor flexibility